

Lecture 9

A Control Perspective on Certifiably Robust Neural Networks, Part III

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9.1 Review: Generalized Lure System

We start from a generalized version of the Lur'e system representation:

$$x' = Ax + B\Delta(Cx + b) + d. \quad (9.1)$$

Here $x \in \mathbb{R}^{n_0}$, $x' \in \mathbb{R}^{n_2}$, and the input to Δ has n_1 entries, so $A \in \mathbb{R}^{n_2 \times n_0}$, $B \in \mathbb{R}^{n_2 \times n_1}$, and $C \in \mathbb{R}^{n_1 \times n_0}$. This slightly generalizes the original Lur'e system $x_{k+1} = A_k x_k + B_k \Delta(C_k x_k)$ by allowing the bias terms (b, d) .

For (9.1), the following theorem holds (you should be familiar with this result).

Theorem 9.1. Assume Δ satisfies

$$\begin{bmatrix} u - v \\ \Delta(u) - \Delta(v) \end{bmatrix}^\top \Lambda \begin{bmatrix} u - v \\ \Delta(u) - \Delta(v) \end{bmatrix} \geq 0, \forall u, v \in \mathbb{R}^{n_1} \quad (9.2)$$

If there exist $\rho \geq 0$ and a positive definite matrix P such that

$$\begin{bmatrix} A^\top P A - \rho^2 P & A^\top P B \\ B^\top P A & B^\top P B \end{bmatrix} + \begin{bmatrix} C & 0 \\ 0 & I \end{bmatrix}^\top \Lambda \begin{bmatrix} C & 0 \\ 0 & I \end{bmatrix} \preceq 0, \quad (9.3)$$

then (9.1) satisfies $(x' - y')^\top P(x' - y') \leq \rho^2(x - y)^\top P(x - y) \forall x, y \in \mathbb{R}^{n_0}$. (Here $x' = Ax + B\Delta(Cx + b) + d$ and $y' = Ay + B\Delta(Cy + b) + d$.)

Proof: From (9.3), we immediately know the following item is non-positive:

$$\begin{bmatrix} x - y \\ \Delta(Cx + b) - \Delta(Cy + b) \end{bmatrix}^\top \left(\begin{bmatrix} A^\top P A - \rho^2 P & A^\top P B \\ B^\top P A & B^\top P B \end{bmatrix} + \begin{bmatrix} C & 0 \\ 0 & I \end{bmatrix}^\top \Lambda \begin{bmatrix} C & 0 \\ 0 & I \end{bmatrix} \right) \begin{bmatrix} x - y \\ \Delta(Cx + b) - \Delta(Cy + b) \end{bmatrix}$$

Since we have $x' - y' = A(x - y) + B(\Delta(Cx + b) - \Delta(Cy + b))$, it is straightforward to verify

$$(x' - y')^\top P(x' - y') = \begin{bmatrix} x - y \\ \Delta(Cx + b) - \Delta(Cy + b) \end{bmatrix}^\top \begin{bmatrix} A^\top P A & A^\top P B \\ B^\top P A & B^\top P B \end{bmatrix} \begin{bmatrix} x - y \\ \Delta(Cx + b) - \Delta(Cy + b) \end{bmatrix}$$

In addition, from (9.2), we have

$$\begin{aligned} & \begin{bmatrix} x - y \\ \Delta(Cx + b) - \Delta(Cy + b) \end{bmatrix}^\top \begin{bmatrix} C & 0 \\ 0 & I \end{bmatrix}^\top \Lambda \begin{bmatrix} C & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} x - y \\ \Delta(Cx + b) - \Delta(Cy + b) \end{bmatrix} \\ &= \begin{bmatrix} (Cx + b) - (Cy + b) \\ \Delta(Cx + b) - \Delta(Cy + b) \end{bmatrix}^\top \Lambda \begin{bmatrix} (Cx + b) - (Cy + b) \\ \Delta(Cx + b) - \Delta(Cy + b) \end{bmatrix} \geq 0 \end{aligned}$$

Putting everything together, we have $(x' - y')^\top P(x' - y') - \rho^2(x - y)^\top P(x - y) \leq 0$, which is the desired conclusion. ■

9.2 Deriving LipSDP for Two-Layer Neural Networks

Consider a simple two-layer neural network:

$$f(x) = W_2 \sigma(W_1 x + b_1) + b_2 \quad (9.4)$$

where σ is the ReLU activation function. We seek a Lipschitz bound $L \geq 0$ such that

$$\|f(x) - f(y)\| \leq L\|x - y\| \quad \text{for all } x, y. \quad (9.5)$$

All norms in this lecture are Euclidean.

We can almost directly view (9.3) as a special case of the generalized Lure system (9.1) by setting $A = 0$, $B = W_2$, $C = W_1$, $\Delta = \sigma$, $b = b_1$, and $d = b_2$. Then we can directly apply Theorem 9.1 to obtain a Lipschitz bound for (9.3) by choosing $\rho = L$ and $P = I$.

Corollary 9.2. *Assume the ReLU activation function σ satisfies*

$$\begin{bmatrix} u - v \\ \sigma(u) - \sigma(v) \end{bmatrix}^\top \Lambda \begin{bmatrix} u - v \\ \sigma(u) - \sigma(v) \end{bmatrix} \geq 0, \forall u, v$$

If the following matrix inequality holds

$$\begin{bmatrix} -L^2 I & 0 \\ 0 & W_2^\top W_2 \end{bmatrix} + \begin{bmatrix} W_1 & 0 \\ 0 & I \end{bmatrix}^\top \Lambda \begin{bmatrix} W_1 & 0 \\ 0 & I \end{bmatrix} \preceq 0,$$

then the two-layer neural network (9.3) is L -Lipschitz, i.e. (9.5) holds.

Now we need to specify Λ for ReLU so the above result can become useful. Notice that ReLU is sloped restricted between $[0, 1]$ (i.e. take any two points on the graph of ReLU and their slope is in the range of $[0, 1]$). Based on this property, for i -th activation entry, we gave $(\sigma(u_i) - \sigma(v_i))((u_i - v_i) - (\sigma(u_i) - \sigma(v_i))) \geq 0$. You can also verify this based on the signs of (u_i, v_i) , e.g. if $u_i \geq 0$ and $v_i \geq 0$, we have $\sigma(u_i) - \sigma(v_i) = u_i - v_i$, and hence $u_i - v_i - (\sigma(u_i) - \sigma(v_i)) = 0$. (Please verify the other three cases yourself).

Let T be a diagonal matrix whose diagonal entries are all positive. Let $T(i, i)$ denote the (i, i) -th entry of T . Then we have

$$\begin{bmatrix} u - v \\ \sigma(u) - \sigma(v) \end{bmatrix}^\top \begin{bmatrix} 0 & T \\ T & -2T \end{bmatrix} \begin{bmatrix} u - v \\ \sigma(u) - \sigma(v) \end{bmatrix} = \sum_i 2T(i, i)(\sigma(u_i) - \sigma(v_i))((u_i - v_i) - (\sigma(u_i) - \sigma(v_i))) \geq 0, \quad (9.6)$$

Therefore, we can choose $\Lambda = \begin{bmatrix} 0 & T \\ T & -2T \end{bmatrix}$, and this leads to the following result.

Corollary 9.3. *If there exists a positive definite diagonal matrix T such that the following matrix inequality holds*

$$\begin{bmatrix} -L^2 I & 0 \\ 0 & W_2^\top W_2 \end{bmatrix} + \begin{bmatrix} W_1 & 0 \\ 0 & I \end{bmatrix}^\top \begin{bmatrix} 0 & T \\ T & -2T \end{bmatrix} \begin{bmatrix} W_1 & 0 \\ 0 & I \end{bmatrix} = \begin{bmatrix} -L^2 I & W_1^\top T \\ TW_1 & -2T + W_2^\top W_2 \end{bmatrix} \preceq 0,$$

then the two-layer neural network (9.3) is L -Lipschitz, i.e. (9.5) holds.

Given (W_1, W_2) , the above condition is linear in L^2 and T , and hence its feasibility can be checked using semidefinite programming (SDP) solvers. For fixed weights (W_1, W_2) , we can find a Lipschitz bound by solving

$$\begin{aligned} & \underset{\zeta, T}{\text{minimize}} && \zeta \\ & \text{subject to} && \zeta \geq 0, \quad T \succeq 0 \text{ diagonal}, \\ & && \begin{bmatrix} -L^2 I & W_1^\top T \\ TW_1 & -2T + W_2^\top W_2 \end{bmatrix} \preceq 0 \end{aligned} \quad (9.7)$$

The constraints are affine in the decision variables ζ and the diagonal entries of T , so this is an SDP. Every feasible solution gives the certified bound $\sqrt{\zeta}$. The optimized certificate is an upper bound on the true Lipschitz constant; the sufficient condition need not be exact.

9.3 LipSDP for Multi-Layer Feedforward Networks

Now we extend LipSDP to address multi-layer feedforward neural networks. Let's start with a three layer network:

$$f(x) = W_3 \sigma(W_2 \sigma(W_1 x + b_1) + b_2) + b_3 \quad (9.8)$$

We can generalize Corollary 9.2 as follows.

Corollary 9.4. *If there exist positive definite diagonal matrices T_1 and T_2 such that the following matrix inequality holds*

$$\begin{bmatrix} -L^2 I & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & W_3^\top W_3 \end{bmatrix} + \begin{bmatrix} W_1 & 0 & 0 \\ 0 & I & 0 \end{bmatrix}^\top \begin{bmatrix} 0 & T_1 \\ T_1 & -2T_1 \end{bmatrix} \begin{bmatrix} W_1 & 0 & 0 \\ 0 & I & 0 \end{bmatrix} + \begin{bmatrix} 0 & W_2 & 0 \\ 0 & 0 & I \end{bmatrix}^\top \begin{bmatrix} 0 & T_2 \\ T_2 & -2T_2 \end{bmatrix} \begin{bmatrix} 0 & W_2 & 0 \\ 0 & 0 & I \end{bmatrix} \preceq 0,$$

then the three-layer neural network (9.3) is L -Lipschitz, i.e. (9.5) holds.

The proof is quite similar to the proof for the previous results. You will be asked to provide a detailed proof for the above result in HW1. The above matrix inequality can be simplified as follows:

$$\begin{bmatrix} -L^2 I & W_1^\top T_1 & 0 \\ T_1 W_1 & -2T_1 & W_2^\top T_2 \\ 0 & T_2 W_2 & W_3^\top W_3 - 2T_2 \end{bmatrix} \preceq 0. \quad (9.9)$$

We can use a very similar proof to obtain the following general LipSDP for neural networks with any depth:

$$\begin{bmatrix} -L^2 I & W_1^\top T_1 & 0 & \cdots & 0 \\ T_1 W_1 & -2T_1 & W_2^\top T_2 & \ddots & \vdots \\ 0 & T_2 W_2 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & -2T_{\ell-1} & W_\ell^\top T_\ell \\ 0 & \cdots & 0 & T_\ell W_\ell & W_{\ell+1}^\top W_{\ell+1} - 2T_\ell \end{bmatrix} \preceq 0. \quad (9.10)$$

Exercise. Write out the 4×4 block linear matrix inequality for $f(x_0) = W_4 \sigma(W_3 \sigma(W_2 \sigma(W_1 x_0 + b_1) + b_2) + b_3) + b_4$.

References

- [1] M. Fazlyab, A. Robey, H. Hassani, M. Morari, and G. J. Pappas. Efficient and accurate estimation of Lipschitz constants for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.